

A RESEARCH PROGRAM, AND WHERE PREDICTIVE CODING FITS IN IT

Generalized Transformers

Compression-driven structure, settling as inference, and caps that keep learning after deployment

Ben Goertzel and colleagues · September 2026

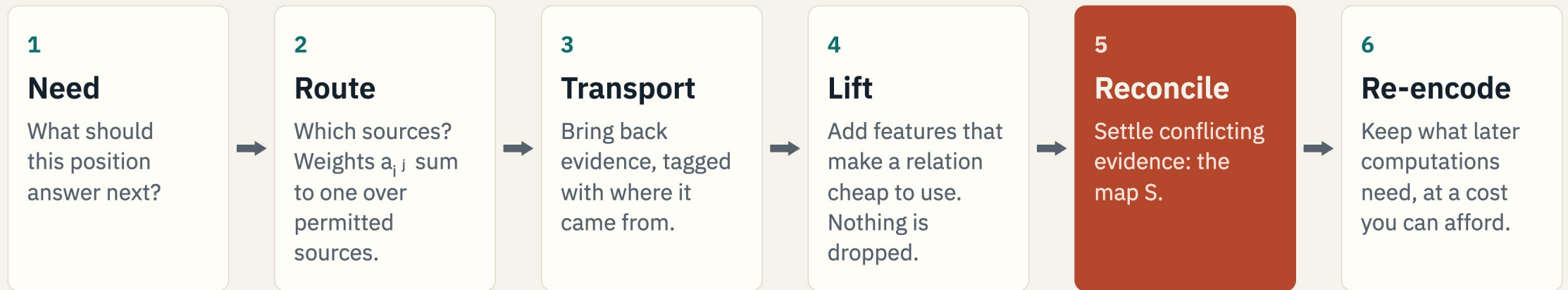
What fitting a fixed architecture leaves out

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|--------------------------------------|--|
| 1 Continual learning | Learn a new domain and keep the old skill. Storing knowledge and still being able to reach it are separate things. |
| 2 Learning during use | Predict, adjust state, update weights and find new structure together, each on its own timescale. |
| 3 Exchange of capabilities | Hand a learned skill to another system as a small module that can be tested before it is trusted. |
| 4 Neural–symbolic links | Let a reasoner or a program supply typed evidence to the same workspace the network reads. |
| 5 Creativity off-distribution | Build and test a new composition when the familiar computations fail. |

A transformer layer does six jobs. Attention plus an MLP is one way of doing them.

Name the jobs and the alternatives become design choices: lookups, small programs, explicit records, and an inference step that settles.

One layer, six jobs



$$x_i^+ = R(S(\Phi[x_i , \{ \sum_j a_{ij} \cdot \delta(j, V(x_j)) \}]))$$

A standard block is the special case: jobs 1–3 are attention with an immediate average, 4 is the MLP, 6 is residual plus norm, and **5 is the identity**. Job 5 is where predictive coding enters: a finite settling map over the lifted state.

Averaging too early loses the binding

“The captain is Mira. The cook is Leo. The captain is _____”

Memory A

captain – Mira
cook – Leo

Memory B

captain – Leo
cook – Mira

Uniform average of the records

Encode each record as one-hot role plus one-hot name and average. A and B give the same vector. Which roles and names occur survives; who holds which is gone, and no later nonlinearity can bring it back.

The binding survives if the layer routes selectively, or lifts a joint role-and-name feature **before** it aggregates. In the framework, evidence arrives as a weighted collection that remembers its sources, and taking the barycenter is a choice made in the open.

What a compressed state must keep

$$d_C(z, z') = \sup \text{ over continuations } c \text{ of } TV(p_c(\cdot \mid z), p_c(\cdot \mid z'))$$

Two states may share one code exactly when no declared later computation can tell them apart.

The price of merging

Give two states at behavioral distance δ one code, and some later computation errs by at least $\delta/2$, whatever the decoder.

Precision from later use

Locally the cost of moving the state is $\frac{1}{2} \Delta^T G(z) \Delta$, with G the Fisher matrix of later predictions. Using it to weight PC errors is untried.

CD as a GT layer

Its re-encoding approximates this quotient: reusable rules, the exceptions they miss, and an index, under memory and work budgets.

Three ways to respond to an error

Revise the answer

Temporary activities settle against the evidence at hand. Perhaps the copy reader deserves more trust on this prefix.

Predictive-coding inference. Fast, every token.

Revise the habit

Persistent weights change after outcomes arrive. Perhaps the gate should generally trust that reader in contexts like this.

PC local rule, or BP on the same computation.

Revise the machinery

Add, remove or compose readers and features; refit; keep what measurably helps. A record of past searches proposes where to look next.

CD structural search. Slow, in the background.

PC already separates the first two. CD adds the third, and persistent structured residuals from settling are a natural signal for it. All of it can sit in a **cap** on a frozen transformer, so the new machinery only has to improve on a model that already works.

The energy view, in familiar terms

$$E_{\theta}(z) = \sum_v \frac{1}{2} \| z_v - f_v(z_{pa(v)}; \theta_v) \|^2 \Lambda_v + E_{\text{evidence}}(z; x)$$

$$E_{\beta}(z) = E_{\theta}(z) + \beta \cdot C(z, y)$$

$$\nabla_{\theta} C(z_{\theta}) = \lim_{\beta \rightarrow 0} [\partial_{\theta} E_{\theta}(z_{\beta}) - \partial_{\theta} E_{\theta}(z_{\theta})] / \beta$$

Free phase predicts

Evidence terms use only what is observed at prediction time. The target enters the nudged phase alone, so scores are never contaminated.

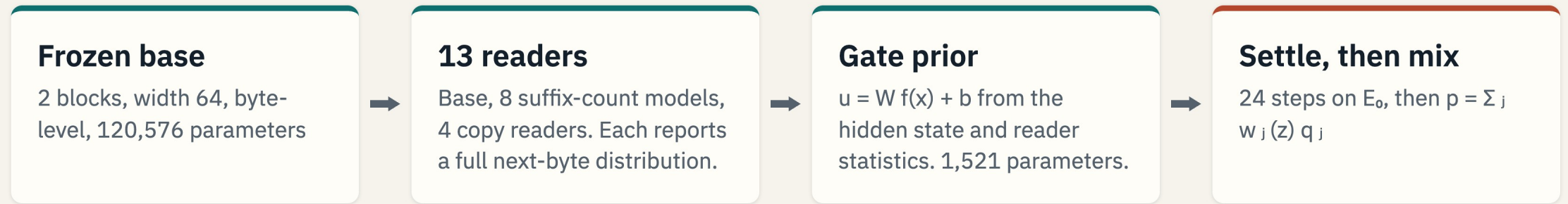
The graph is not fixed

CD may propose new nodes, edges and readers. After each structural change the affected parameters are refitted and compared with a control.

A matched BP control

CD-BP differentiates through the same K settling steps. Inference is identical in both arms, so a difference can be credited to the learning rule.

H2: a small cap on a frozen transformer



$$E_{\theta}(z ; x) = \frac{1}{2} \| z - u_{\theta}(x) \|^2 + \lambda \cdot (1/m) \sum_s -\log \sum_j w_j(z) q_{j,s}$$

$q_{j,s}$ is what reader j predicted for the byte that then occurred at past step s , over the last $m = 16$ bytes. The learned prior proposes trust; recent prediction errors revise it. CD search chooses which 5 of the 13 readers stay active, by fitting all and pruning.

Where the gain comes from

2.90 → 2.34

bits per byte on held-out Shakespeare, five-seed mean. The controls below say which part of the system earns which part of the gain.

Condition	bpb	What it isolates
Frozen transformer	2.8962	the base
Global mixture of all 13 readers, no gate	2.4285	richer evidence does most of the work
Fixed five readers, PC, 1,056 updates	2.3460	a learned, context-sensitive gate
Compression-first CD–PC, five readers	2.3357	structure search: 0.0102 better, 5/5 seeds
All 13 readers, PC	2.3323	the accuracy reference

What predictive coding did, and did not do

0.0008

bpb by which the matched CD–BP arm beat CD–PC. As a learning rule PC ties with BP here, and its updates track BP gradients as β shrinks and settling lengthens.

+0.0025

bpb lost when settling is switched off at prediction time, in 5 of 5 seeds. Feeding it another example's recent evidence costs +0.0023, so the benefit comes from the evidence and not from extra compute.

+0.0048

extra forgetting on Shakespeare after adapting to a new domain (Alice) when settling is switched off: 0.0660 becomes 0.0708. Inference gives some protection to old skills.

So far PC earns its place here as **inference**: reconciling trust in readers with recent evidence. The effects are small and consistent, on a gate with 1,521 parameters. Whether they grow with richer graphs and real hierarchies is the open question, and it is a PC question.

Forgetting under domain shift

Share of the cap's Shakespeare advantage erased after 1,024 updates on the new domain



Plasticity is kept

With replay the cap still gets 97.7% and 99.3% of the new-domain gain.

PC and BP curves coincide

The learning rule makes no visible difference to forgetting.

Readers and dictionary were frozen; only the gate moved. The knowledge stayed **available** and **access** to it drifted: the library kept its books while the librarian picked up new habits.

Open problems where PC expertise counts

Caps as summed energies

$E_A = E_B + \sum_j \alpha_j E_j$ over active modules.
A unique minimum? Does settling order change the answer? How to train subsets to cooperate?

Precision from later use

Weight state errors by the Fisher matrix of later predictions, so the energy attends to directions that change behavior.

Residuals guide search

Persistent structured errors suggest a missing observer. Rank candidates by $\frac{1}{2} g^T (H + \lambda I)^{-1} g$, then refit against a control.

Affordable settling

Warm starts, a learned predictor of the settled state, settling only when uncertain. In the first cap PC ran several times slower than BP.

Locality and retention

Do local updates let protected, versioned modules grow without disturbing old routing? Only replay is measured so far.

Learning while serving

Fast settling, slower local updates, background structure search, atomic version publishing. Designed, not yet benchmarked.

A learner that can preserve, reach, revise and share structure while it runs

Now

Replicate H2 on fresh small corpora. Growing readers, retained gates, strict memory ceilings, longer task streams.

Next

A 10M–100M base on one accelerator, against a strong ordinary adapter at matched memory and compute.

Then

A 0.1B–3B open base with composable caps, symbolic readers, and held-out compositions as a creativity test.

Later

Module exchange between workers with different data; a frontier-base cap; a standalone GT/CD learner only if the smaller steps hold.

The full report, with proofs, algorithms and the evidence bundle behind every number, is available on request.